

# PAX path stability — run report

## RUN

174506\_pax-path-hold\_pax-path-hold

Status: completed | Started: 2026-09-23T16:45:06.796904+00:00

## PURPOSE

Hold both outputs at zero and record PAX/PD for each manual path condition. Duration is per condition.

## REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted. Photodiode connected to configured RP input.

## RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

## OPERATOR SETUP NOTES

V\_pi expectation only; verify channel and measured driver gain before calibration.

## DATA

723 recorded samples; path\_a\_only: 241, path\_b\_only: 241, both\_paths: 241

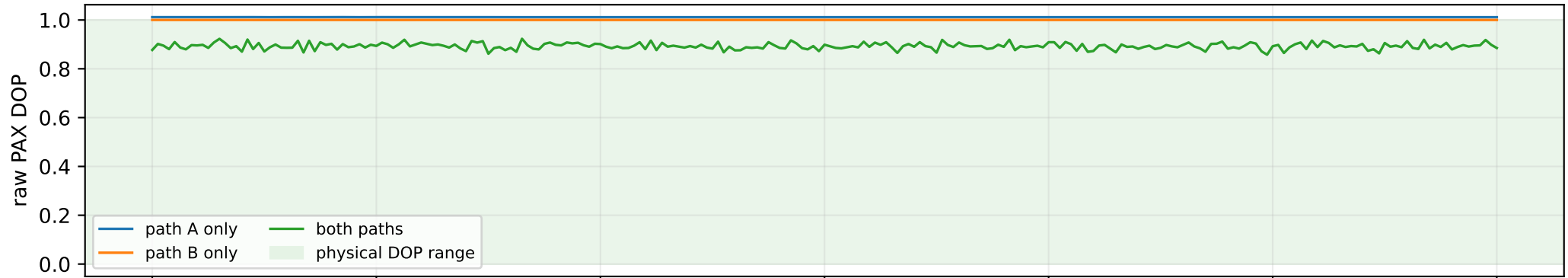
Source: data.csv; full acquisition settings: recipe.json; status: run.json

## RAW DOP

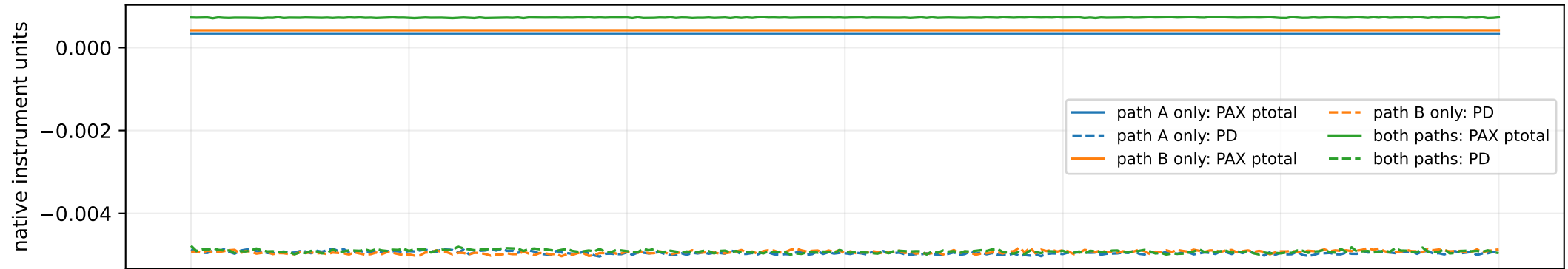
Mean 0.9682; range 0.8576 to 1.0119. 470 samples outside 0-1. Telemetry is not a new calibration.

# Fixed-state PAX path diagnostic — 174506\_pax-path-hold\_pax-path-hold

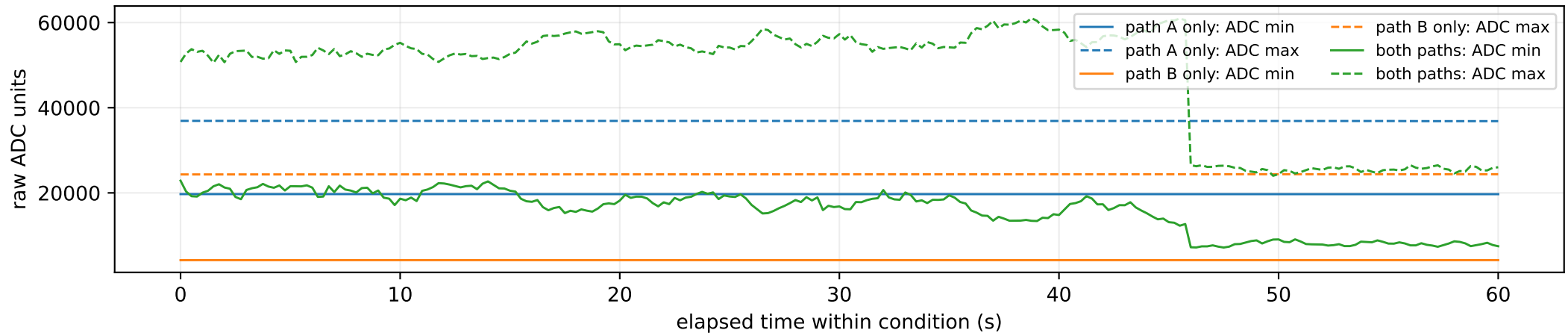
Raw DOP while actuator commands are fixed at 0 V



Concurrent, uncalibrated power telemetry



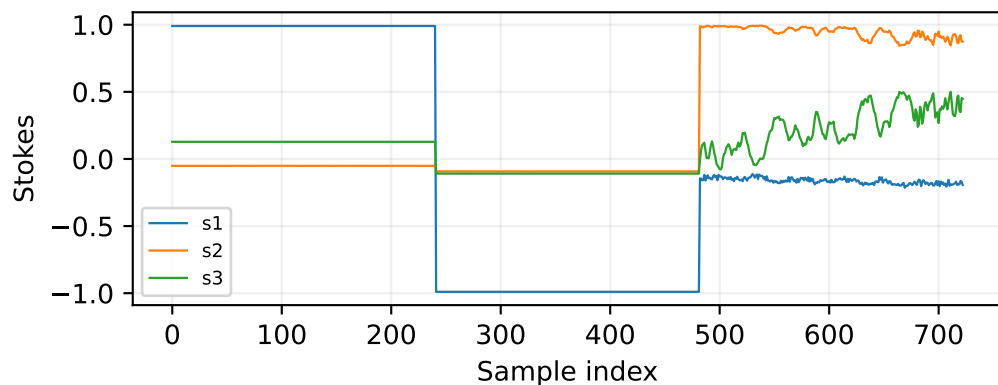
PAX primary-record ADC extrema



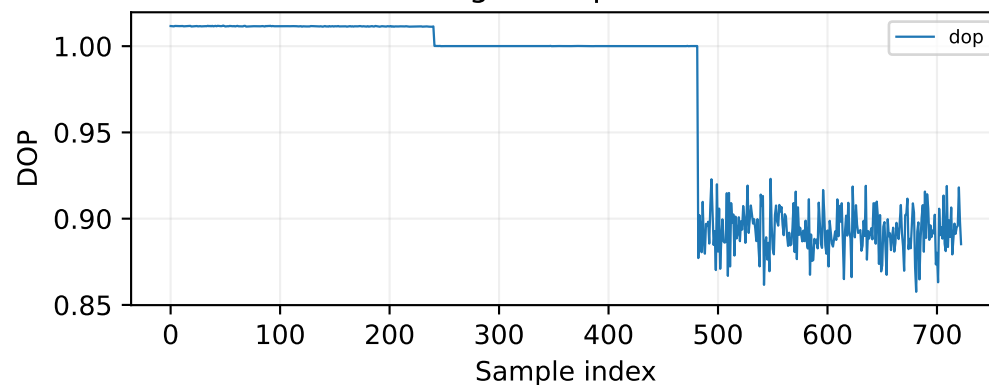
# Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

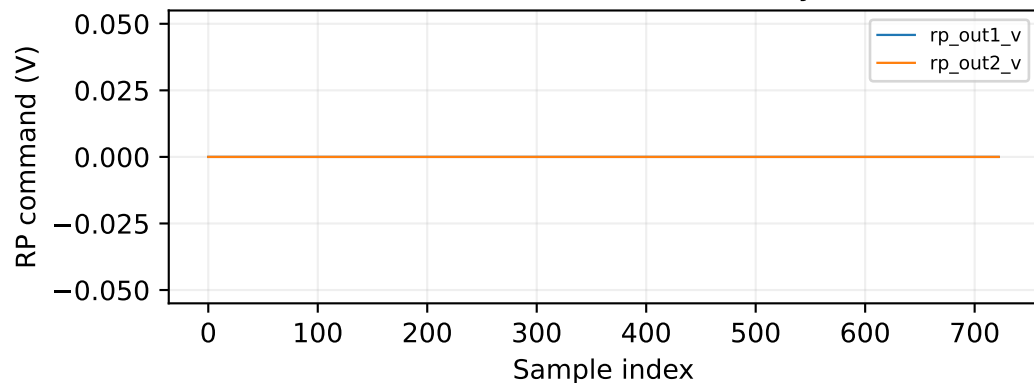
## Normalized Stokes



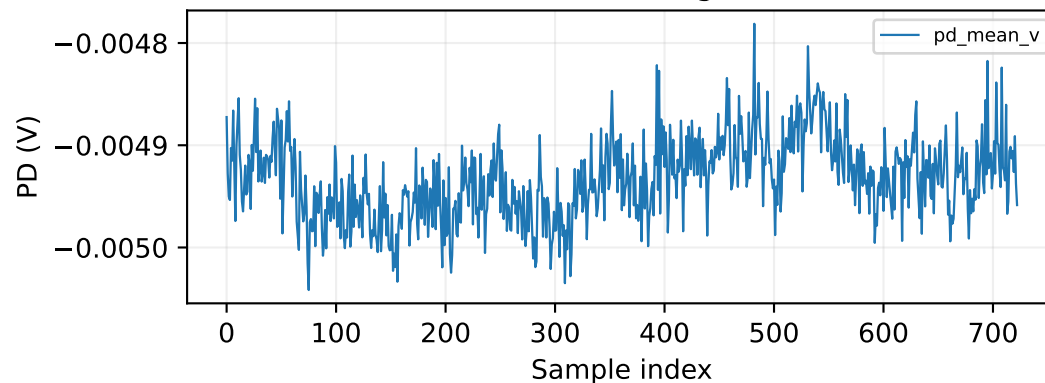
## Raw degree of polarization



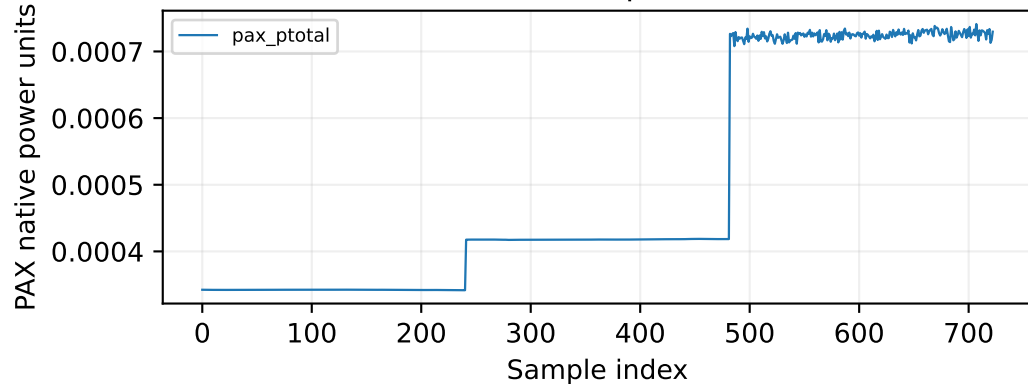
## Actuator command history



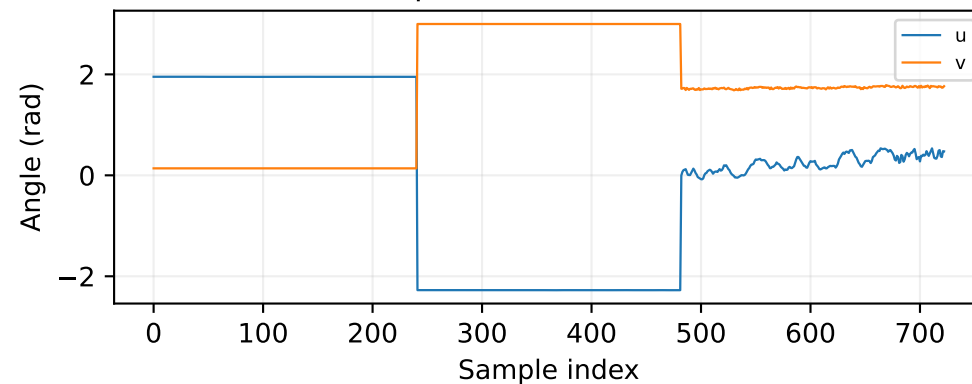
## Photodiode signal



## PAX total power



## Sphere coordinates



# Active parameters

RP commands: volts, bounded to 0–1. V\_lambda: terminal volts for 2π. Gains: terminal V / RP V.

| Parameter                       | Recorded value  |
|---------------------------------|-----------------|
| label                           | "pax-path-hold" |
| duration_s                      | 60.0            |
| report                          | "pdf"           |
| pax_rotation_velocity_hz        | 60.0            |
| pax_measurement_mode            | 5               |
| pax_wavelength_nm               | 830.0           |
| phi1_v_lambda                   | 12.2            |
| phi2_v_lambda                   | 12.2            |
| phi1_actuator_volts_per_rp_volt | 16.875          |
| phi2_actuator_volts_per_rp_volt | 16.875          |
| rp_output_min_voltage           | 0.0             |
| rp_output_max_voltage           | 1.0             |
| pd_input                        | "in1"           |
| pd_nd_optical_density           | 2.0             |
| pd_scope_duration_s             | 0.01            |
| pd_scope_decimation             | 64              |
| pd_scope_timeout_s              | 2.0             |
| pax_path_hold_sample_period_s   | 0.25            |